

Q9			
(a)	(i)	As the coil rotates at constant angular speed ω, the component of magnetic flux density B perpendicular to the cross-sectional area A of the coil varies sinusoidally. Since the magnetic flux linkage Φ varies sinusoidally, the rate of change of flux linkage will be sinusoidal. For instance, if $\Phi = NBA \sin \omega t$, then $d\Phi/dt = NBA \omega \cos \omega t$. By Faraday's law, the magnitude of the induced emf is directly proportional to the rate of change of magnetic flux linkage. Thus emf induced in the coil will be sinusoidal.	
	(ii)	1. $(6.8 \text{ cm} \times 0.050 \text{ V cm}^{-1})/2$ $= 0.17 \text{ V}$ 2. $T = 5 \text{ cm} \times 8.0 \text{ ms} = 40 \text{ ms}$ $f = 1/(40 \times 10^{-3}) = 25 \text{ Hz}$	
	(iii)	The plane of the rectangular coil is perpendicular to the plane of the paper (i.e. minimum flux linkage), when maximum e.m.f. is induced in the coil. OR coil rotated 90°	
	(iv)	$E_0 = NBA \omega$ $0.17 = 120 \times B \times 0.0013 \times 2\pi(25)$ $B = 6.94 \times 10^{-3} \text{ T}$	
(b)	(i)	The root-mean-square value of an alternating current is the equivalent constant direct current that will dissipate the same amount of heat per unit time in a given resistive load.	
	(ii)	$\omega = 2\pi f = 377$ $f = \frac{377}{2\pi} = 60.0 \text{ Hz}$	
	(iii)	$V_{rms} = \frac{V_o}{\sqrt{2}} = \frac{240}{\sqrt{2}}$ Power dissipated in heater, $\langle P \rangle = \frac{(V_{rms})^2}{R}$ $= \left(\frac{240}{\sqrt{2}} \right)^2 \left(\frac{1}{38} \right)$ $= 758 \text{ W}$	

- (iv) The peak power, $P_o = (240)^2/38$
 $= 1516 \text{ W}$

